Mini Project Report

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Sub: Advanced Programming Concepts

Div: B Batch:T1

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1. Domain

Artificial Intelligence / Expense Management

This project is related to AI-based expense management. It helps users scan paper receipts using OCR technology, automatically extracts important details, and stores expenses digitally for easy tracking.

2. Project Title

TraceLens Lite – Receipt Scanner & Expense Tracker Using Python

3. Problem Statement

Many people still keep paper receipts for shopping, food, travel, and other expenses. These receipts can be lost, damaged, or forgotten, making it difficult to track spending. Manually entering expenses into a notebook or Excel is time-consuming.

The proposed system allows users to upload a receipt image, extracts details such as shop name, date, total amount, and purchased items using OCR, and stores the data in a digital dashboard for future reference and expense analysis.

4. Requirements / Technology

Software Requirements

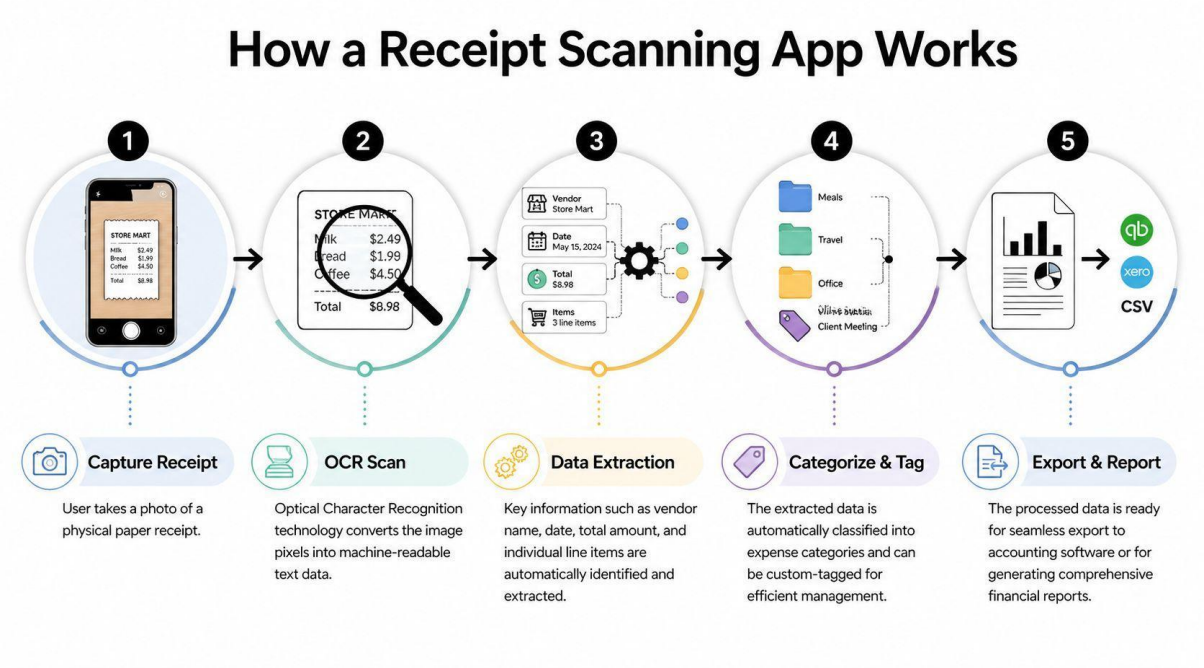
1. Programming Language: Python
2. Web Framework: Flask
3. OCR Library: EasyOCR
4. Database: SQLite
5. Frontend: HTML, CSS, JavaScript
6. Code Editor: Visual Studio Code

Hardware Requirements

1. Processor: Intel i3 or above

2. RAM: Minimum 4 GB
3. Storage: Minimum 500 MB free space
4. Internet connection (optional for installing libraries)

5. System Architecture



Working of the Project

1. User uploads a receipt image.
2. The image is processed and enhanced.
3. EasyOCR extracts text from the receipt.
4. Important details (shop, date, amount, items) are identified.
5. The expense is saved in the SQLite database.
6. The dashboard displays expense history and records.